CO-2 ASSESSMENT TOOL-3

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Course: Machine Learning

Course Code: ITA0612

Slot: A

1. A convolutional neural network (CNN) used for image classification produces unstable results across training epochs. Examine potential issues such as improper data augmentation, vanishing gradients, or batch normalization errors. Describe debugging strategies and suggest optimization techniques to stabilize training and improve accuracy.

Ans:

A Convolutional Neural Network (CNN) is used for image Classification. Unstable results across training epochs may occur due to data, gradient, or normalization problems.

Potential Issues:

1. Improper data augmentation: Excessive or incorrect transformations may distort images and reduce accuracy.
2. Vanishing gradients: Gradients become very small, preventing early layers from learning effectively.
3. Batch normalization errors: Incorrect training/evaluation mode or unsuitable batch sizes may cause unstable Predictions.

Debugging and optimization:

1. Check image labels, preprocessing, and augmented images.
2. Monitor training and validation loss to identify instability and overfitting.
3. Use ReLU or suitable activation functions and proper weight initialization to reduce vanishing gradients.
4. Correctly configure batch normalization and use an appropriate batch size.
5. Apply learning-rate scheduling, Adam optimization and early stopping.

2. A reinforcement learning agents fails to learn an optimal policy in a simulated environment. Investigate causes such as poor reward design, insufficient exploration, or incorrect state representation. Explain debugging methods and recommend improvements like reward shaping, exploration strategies and tuning hyperparameters.

Ans:

Reinforcement Learning (RL) is a machine learning technique in which an agent learns by interacting with an environment and receiving rewards or penalties.

Potential issues:

1. Poor reward design: Rewards do not encourage the desired behavior.
2. Insufficient exploration: The agent repeatedly chooses familiar actions and misses better alternatives.
3. Incorrect state representation: Important environmental information is missing.
4. Improper hyperparameters: Learning rate or discount factor may prevent effective learning.

Debugging and optimization:

1. Monitor cumulative rewards and episode length during training.
2. check whether the state, action, reward and next-state values are correct.
3. Use reward shaping to provide useful intermediate rewards without changing the intended optimal policy.
4. Apply exploration strategies such as epsilon-greedy or entropy-based exploration.
5. Tune the learning rate, discount factor, and exploration rate; compare performance across multiple runs.

A clustering algorithm like K-means produces inconsistent cluster assignments for similar datasets. Analyze possible reasons such as poor initialization, incorrect number of clusters, or sensitivity to outliers. Propose debugging approaches and optimization strategies like K-means++, normalization, and silhouette analysis.

Ans:

K-means is an unsupervised learning algorithm that groups data points into K clusters based on their distance from cluster centroids.

Potential Issues:

1. Poor initialization: Different initial centroids produce different clusters.
2. Incorrect value of K: Too few or too many clusters lead to poor grouping.
3. outliers: Extreme values shift centroids away from normal data points.
4. Unscaled features: Features with large numerical ranges dominate distance calculations.

Debugging and optimization:

1. Check missing values, duplicate records, and outliers.
2. Normalize or standardize features before clustering.
3. Use K-means++ to select better initial centroids.
4. Run the algorithm multiple times with different initializations.
5. Use the Elbow Method and Silhouette Score to select a suitable number of clusters and evaluate cluster quality.

Result:

Better initialization, feature scaling, and selection of K improve clustering consistency and quality.

4. A Support vector machine (SVM) model underperforms on a nonlinear dataset. Identify issues such as incorrect kernel choice, improper parameter tuning, or scaling problems. Suggest debugging techniques and optimization methods including kernel selection, hyperparameter tuning, and feature scaling.

Ans:

A Support Vector Machine (SVM) is a supervised learning algorithm that finds a decision boundary separating different classes. It can handle nonlinear data using kernel functions.

Potential Issues:

1. Incorrect Kernel: A linear kernel may fail to separate nonlinear data.
2. Poor hyperparameter tuning: Incorrect values of C and gamma may cause underfitting or overfitting.
3. Feature scaling problems: Features with larger ranges may dominate the decision boundary.
4. Noisy or irrelevant features: These may reduce classification performance.

Debugging and Optimization:

1. Inspect class distribution, missing values, and incorrect labels.
2. Apply StandardScaler to normalize feature scales.
3. Try linear, polynomial, and RBF kernels depending on the dataset.
4. Use Grid Search or Random Search with cross-validation to tune C, gamma and kernel parameters.
5. Evaluate accuracy, precision, recall and F1-score on unseen test data.

5. A natural language Processing (NLP) model using recurrent neural networks (RNNs) suffers from poor long-term dependency learning. Examine causes such as vanishing gradients and limited memory capacity. Describe debugging steps and propose solutions like LSTM/GRU units, attention mechanisms, and improved training strategies.

Ans:

A Recurrent Neural Network (RNN) processes sequential data, such as sentences, by maintaining information from previous time steps. It may struggle to learn long-term dependencies.

Potential Issues:

1. Vanishing gradients: Gradients become very small during backpropagation through long sequences.
2. Limited memory: A basic RNN may forget important information from earlier words.
3. Long sequences: Long inputs make training difficult and computationally expensive.
4. Insufficient training: Poor learning rates or inadequate training data may reduce performance.

Debugging and optimization:

1. Check tokenization, padding, sequence lengths and input labels.
2. Monitor training and validation loss to identify learning problems.
3. Use LSTM or GRU units, which use gates to retain important information over longer sequences.

4. Apply attention mechanisms so the model can focus on relevant words in the input sequence.
5. Use gradient clipping, suitable learning rates, and appropriate sequence lengths to stabilize training.

Result: LSTM/GRU networks, attention mechanisms and improved training strategies help capture a long-term dependencies and improve NLP performance.